AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M110
Python Programming

Midterm Examination - (MTA)
Fall Semester 2025/2026

Date: 19/Nov./2025

Number of Exam Pages: 3 (including this cover sheet)	Time Allowed: 2 Hours

Instructions:

- Total Marks: **30**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: Multiple Choices questions [5 Marks]

Question1: Answer the following questions by choosing the most correct statement.

1- All keywords in Python are ____
   a. Capitalized
   b. In lower case
   c. In upper case
   d. None of the choices

2- Which of the following is used to define a block of code in Python language?
   a. Brackets
   b. indentation
   c. Parentheses
   d. All the mentioned

3- Which of the following character(s) is used to give single-line comments in Python?
   a. //
   b. //*
   c. #
   d. Both a and c.

4- Which part of the expression **(2 ** (2 + 6)/ 3 *2)** is evaluated first?
   a. 3*2
   b. (2+6)/3
   c. 2**
   d. (2+6)

5- What arithmetic operator(s) cannot be used with strings in Python?
   a. +
   b. -
   c. *
   d. Both b and c.

PART 2: Problem Solving Questions [25 Marks]

Question 2: [6 marks]
What is the output of the following programs?

a- (2 marks)

```
x = 10
while x > 0:
    if x % 3 == 0:
        print(x, end=" ")
    x -= 2
print("\nEnd!")
```

b- (2 marks)

```
for i in range(4, 0, -1):
    for j in range(i):
        print(i, end="")
    print()
```

c- (2 marks)

```
nums = [2, 5, 3, 7]
nums.append(1)
nums.sort()
nums.reverse()
print(nums)
print(nums[1:4])
```


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Question 3: [4 marks]
Draw the flowchart that represents the logic of a program that repeatedly asks the user to enter positive numbers. The program should keep a running total of all entered numbers and count how many numbers were entered.
The program stops when the user enters a negative number.
Finally, the program should display:

- The total sum of the entered numbers
- The average of the entered numbers

Question 4: [7 marks]
Write a Python program that prompts the user to enter a **letter grade** (A–F).
The program should determine the **corresponding GPA value** using **chained if-elif-else statements**. (Assume that the entered value is an uppercase letter.)

- If the input is valid (within the range A–F), the program should assign the corresponding GPA value.
- If the input is invalid (not within the range A–F), the program should assign the value **-1** to the GPA.
- At the end of the program, print the GPA value if it is not equal to -1; otherwise, display an appropriate error message.

Grade conversions:

- *A : 4.0*
- *B : 3.0*
- *C : 2.0*
- *D : 1.0*
- *F : 0.0*

Example:

- If the input is 'B', the program should print: "The GPA value for grade B is 3.0."
- If the input is 'G', the program should print: "Invalid grade entered."

Question 5: [8 marks]
Write a Python program that performs the following operations on the list:
temperatures = [22, 25, 19, 30, 27]

1. **Iteration (1 mark):** Print all temperature values separated by spaces.
2. **Find Maximum (2 marks):** Determine and print the highest temperature in the list.
3. **Update a Value (2 marks):** Prompt the user to input an index and a new temperature.
    - If the index is valid, update the temperature at that index.
    - Otherwise, display an error message.
4. **Add a New Temperature (2 marks):** Prompt the user to input a new temperature to add to the end of the list.
5. **Display the Final List (1 mark):** Print the final state of the list after all operations.

Sample Output:

```
22 25 19 30 27
The highest temperature is 30
Enter the index to update: 2
Enter the new temperature: 26
Temperature at index 2 has been updated.
Enter a new temperature to add: 31
[22, 25, 26, 30, 27, 31]
```

End of questions

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